AVNISH DESHMUKH

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PROFESSIONAL SUMMARY

Robotics engineer (final-year B.Tech, ECE – AI & ML) with 5 years on the MIT WPU-Tech DD Robocon team across electronics, embedded systems and mechanical design, now working on state estimation for legged robots as a DRDO intern. Designed and debugged custom PCBs and embedded control systems for competition robots; 2+ years of ROS 2 for system integration; 1 year designing lightweight mechanisms in SolidWorks and manufacturing them via 3D printing, laser cutting, milling and turning. Comfortable owning a subsystem end-to-end, from schematic to firmware to mechanical fit.

PROFESSIONAL EXPERIENCE

Research Intern – Quadruped Robotics | DRDO | 2026 – Feb 2027

- State estimation for a legged robot operating in GPS-denied settings.

ACHIEVEMENTS & COMPETITIONS

ABU Robocon / DD Robocon – National Competition

- All India Rank 2 – DD Robocon 2026 | All India Rank 3 – DD Robocon 2024
- All India Rank 2 – DD Robocon 2023 | All India Rank 3 – DD Robocon 2022

TECHNICAL EXPERIENCE – MIT WPU-TECH TEAM (DD ROBOCON)

Embedded Systems & Electronics | 2022 – Present

- Owned electrical and embedded subsystem design across 5 competition cycles (2022–2026), from schematic through firmware to on-robot integration.
- Designed and fabricated 10+ custom PCBs for motor control, sensor interfacing and pneumatic actuation, removing third-party module dependencies and field failure points.
- Programmed and debugged STM32-based control systems handling ~10 actuators and ~15 sensors.

Programming & Robotics Software | 2024 – Present

- Developed embedded software for microcontrollers, including custom libraries and integration of sensors, actuators and communication interfaces.
- Used ROS 2 since 2024 for robotic software development, hardware integration and navigation.

Mechanical Design & Manufacturing | 2026

- Designed robotic mechanisms in SolidWorks and took them from CAD to fabricated, assembled parts using 3D printing, laser cutting, milling, turning and other machining processes.

LEADERSHIP

- **Vice-Captain, MIT WPU-Tech Team** | 2026 – Led a 15-member ABU/DD Robocon team.
- **Technical Mentor, Electronics & Cross-Department Recruits** | 2023 – Present – Mentored ~20 recruits.

EDUCATION

MIT World Peace University (MIT-WPU), Pune | Expected 2027

Integrated B.Tech in Electronics & Communication Engineering (AI & ML) – CGPA 7.36/10

TECHNICAL SKILLS

PCB & Electronics: Altium Designer, Autodesk EAGLE, PCB design, embedded systems, electronics debugging

Programming & Robotics: C, C++, Python, ROS 2, ros2_control, Docker, Gazebo

Mechanical: SolidWorks, mechanical design, manufacturing & fabrication

Motor Control & Hardware: ODrive, VESC, motor drivers, STM32, ESP32

Sensors & Navigation: VectorNav IMUs, GPS, SICK distance sensors, Baumer proximity sensors, rotary encoders

PROJECTS

Robotics Software

- **ODrive ROS 2 Hardware Interface** – ros2_control hardware interface for ODrive controllers, providing motor command and feedback integration.
- **DualShock 4 ROS 2 Interface** – maps DualShock 4 inputs to ROS 2 commands for robot teleoperation.
- **Pure Pursuit Navigation** – autonomous path following in ROS 2 using a Pure Pursuit controller.

Embedded Systems & Electronics

- **ESP32 & STM32 Breakout/Shield Boards** – custom PCBs and firmware for robot control and hardware interfacing.
- **Smart Pneumatic Solenoid Control Board** – custom PCB using protected high-side switch/driver ICs for reliable pneumatic actuation.
- **Robocon Robot Electronics Integration** – integrated controllers, sensors, motor drivers and communication interfaces across multiple competition robots.

Mechanical Design & Robotic Mechanisms

- **Staff Handling & Lifting Robot (prototype)** – sub-25 kg mobile robot with automated staff pickup, assembly and 25 kg load lifting.
- **Basketball Throwing Mechanism** – prototyped and evaluated wheel-based and linear throwers; contributed to calculations, actuator selection, electronics and control.
- **Ring-Throwing Catapult** – BLDC-driven catapult (Nidec gearbox, ODrive); contributed to mechanical and electronics integration.

Final-Year Capstone

- **Autonomous College Guide AGV** – developing an autonomous guided vehicle that works as a campus guide robot.